# Supplementary Material

Where Does LLM-Based Mathematical Verification Still Fail? An Externally Validated Gap Analysis for Reliable AI Tutoring

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**Contents.** S1: the 46-paper corpus with cluster membership. S2: the external-validation log for the ten candidate gaps. S3: the independent second classification of the gap statuses.

## S1. Corpus (46 papers)

Corpus searched in the first week of June 2026 (Consensus as the main tool, plus Google Scholar and Semantic Scholar). *Type* follows the corpus list: 3 surveys, 35 peer-reviewed, 8 preprints. *Clusters* are the limitation clusters C1–C11 of Table 1 in the paper (cluster lists from the Phase 1 extraction); a paper may belong to several clusters, so memberships sum to 70. *Gaps* are the candidate gaps the paper's clusters feed (C1→G1, C2→G4, C3→G5, C4/C11→G2, C5→G7, C6→G3, C7→G8, C8→G10, C9→G9, C10→G6). Twelve papers contribute to no cluster: their limitations were specific to one paper or fell among the isolated issues (fewer than three papers). * = cited in the paper. The last column gives one representative author-stated limitation, paraphrased from the five-field extraction. Venues and years were checked against publisher, ACL Anthology or arXiv pages on 14 October 2026.

| # | Title | First author | Year | Venue | Type | Clusters | Gaps | Representative stated limitation |
|---|---|---|---|---|---|---|---|---|
| 1 | A Survey on Large Language Models for Mathematical Reasoning* | Wang et al. | 2025 | ACM Computing Surveys | Survey | C4, C5, C8, C11 | G2, G7, G10 | Correct answers despite flawed intermediate steps; PRM signals costly and noisy |
| 2 | Multi-Step Reasoning with Large Language Models, a Survey* | Plaat et al. | 2025 | ACM Computing Surveys | Survey | C4, C5, C8, C11 | G2, G7, G10 | Errors accumulate across steps with no mechanism to localise or halt them; self-reflection is prompted, not autonomous |
| 3 | From System 1 to System 2: A Survey of Reasoning LLMs* | Li et al. | 2026 | IEEE TPAMI | Survey | C4, C5, C7, C8, C11 | G2, G7, G8, G10 | Automated step supervision still uses final-answer correctness as proxy oracle |
| 4 | Evaluating Mathematical Reasoning Beyond Accuracy (ReasonEval)* | Xia et al. | 2025 | AAAI 2025 | Peer-reviewed | C4, C7 | G2, G8 | 16-20% false-positive floor in step-level error detection |
| 5 | Evaluating Mathematical Reasoning of LLMs: Error Identification and Correction* | Li et al. | 2024 | Findings of ACL 2024 | Peer-reviewed | C4, C10 | G2, G6 | Only single-step, single-type errors evaluated |
| 6 | Stepwise Verification and Remediation of Student Reasoning Errors with LLM Tutors* | Daheim et al. | 2024 | EMNLP 2024 | Peer-reviewed | C1, C2, C5, C6, C9 | G1, G3, G4, G7, G9 | F1 0.70 → 0.56 without reference solution; model-generated student chains; first error only |
| 7 | LEMMA: Learning from Errors for Mathematical Advancement in LLMs | Pan et al. | 2025 | Findings of ACL 2025 | Peer-reviewed | C1 | G1 | First-error identification relies on GPT-4o annotation |
| 8 | Dissecting Failure Dynamics in Large Language Model Reasoning | Zhu et al. | 2026 | ACL 2026 | Peer-reviewed | C1 | G1 | Segment validity relies on an external oracle model |
| 9 | Is Mathematical Problem-Solving Expertise in LLMs Associated with Assessment Performance? | Zhang et al. | 2026 | HCII 2026 | Peer-reviewed | C1, C2, C9 | G1, G4, G9 | Curated traces, not authentic student work; error-present accuracy 4.7-10.0% |
| 10 | SuperCorrect: Thought Template Distillation and Self-Correction | Yang et al. | 2025 | ICLR 2025 | Peer-reviewed | – | – | Depends on frontier teacher LLMs; locate correctness at most 0.67 |
| 11 | Controlling Equational Reasoning in LLMs with Prompt Interventions | Meadows et al. | 2025 | AAAI 2025 | Peer-reviewed | C10 | G6 | Template detection covers only 6 error categories; interventions affect several error types at once |

| # | Title | First author | Year | Venue | Type | Clusters | Gaps | Representative stated limitation |
|---|---|---|---|---|---|---|---|---|
| 12 | MathOdyssey: Benchmarking Mathematical Problem-Solving Skills in LLMs | Fang et al. | 2025 | Scientific Data | Peer-reviewed | C11 | G2 | Does not assess intermediate reasoning quality; contamination and coverage limits |
| 13 | S3c-Math: Spontaneous Step-Level Self-Correction* | Yan et al. | 2025 | AAAI 2025 | Peer-reviewed | C8 | G10 | Self-correction trained via constructed data; reflection quality not examined |
| 14 | Enhancing Mathematical Reasoning in LLMs by Stepwise Correction (StepCo) | Wu et al. | 2025 | ACL 2025 | Peer-reviewed | – | – | Fails when problem difficulty exceeds the verifier's capacity; English only |
| 15 | AutoPSV: Automated Process-Supervised Verifier* | Lu et al. | 2024 | NeurIPS 2024 | Peer-reviewed | C1, C3, C7 | G1, G5, G8 | Confidence change may not reflect step correctness; task-dependent thresholds |
| 16 | Probabilistic Soundness Guarantees in LLM Reasoning Chains (ARES)* | You et al. | 2025 | EMNLP 2025 | Peer-reviewed | C3, C5 | G5, G7 | Depends on entailment-model calibration; limited propagated-error annotation in real data |
| 17 | Step-by-Step Correction of LLM-based Math Word Problem Solutions* | Li et al. | 2025 | ICASSP 2025 | Peer-reviewed | C10 | G6 | Error types often entangled; detection is the bottleneck |
| 18 | MPBench: Multimodal Reasoning Benchmark for Process Error Identification | Xu et al. | 2025 | Findings of ACL 2025 | Peer-reviewed | – | – | Possible inaccurate error-location labels on hard problems |
| 19 | Teaching LLMs for Step-Level Automatic Math Correction via RL (StepAMC)* | Li et al. | 2025 | ICME 2025 | Peer-reviewed | C1, C2 | G1, G4 | "Lack of real-world datasets containing students' step-by-step solutions" |
| 20 | Improving Human Verification of LLM Reasoning through Interactive Explanation Interfaces | Zhou et al. | 2026 | IUI 2026 | Peer-reviewed | C6 | G3 | Future: adaptive explanations personalised to users |
| 21 | LLMs can Find Mathematical Reasoning Mistakes by Pedagogical Chain-of-Thought (PedCoT)* | Jiang et al. | 2024 | IJCAI 2024 | Peer-reviewed | C6, C9 | G3, G9 | Covers only lower three Bloom levels; static evaluation |
| 22 | A Theoretical Understanding of Chain-of-Thought: Coherent Reasoning and Error-Aware Demonstration | Cui et al. | 2025 | AISTATS 2025 | Peer-reviewed | – | – | Gaussian-input assumption; error explanations created manually |
| 23 | Achieving >97% on GSM8K: Deeply Understanding the Problems (DUP) | Zhong et al. | 2025 | Frontiers of Computer Science | Peer-reviewed | – | – | Three-stage prompting raises inference cost; generality unverified |
| 24 | AutoCrit: A Meta-Reasoning Framework for Self-Critique and Iterative Error Correction | Sang | 2025 | ICMLCA 2025 (IEEE) | Peer-reviewed | C8 | G10 | Relies on the critique agent's reliability; over-correction drift |
| 25 | CHAMP: Competition-level Dataset for Fine-Grained Analyses of LLMs' Math Reasoning* | Mao et al. | 2024 | Findings of ACL 2024 | Peer-reviewed | C4, C11 | G2 | Final-answer accuracy overestimates reasoning ability; 270 problems |
| 26 | Knowledge-Centered Dual-Process Reasoning for Math Word Problems with LLMs | Liu et al. | 2025 | IEEE TKDE | Peer-reviewed | – | – | Verifier rejects valid alternative paths fixed to one annotated expression |
| 27 | Rewarding Graph Reasoning Process makes LLMs more Generalized Reasoners (GraphPRM) | Peng et al. | 2025 | KDD 2025 | Peer-reviewed | – | – | Accuracy degrades with graph size; math PRMs do not transfer |
| 28 | Detecting Hallucinations in LLMs Using Semantic Entropy* | Farquhar et al. | 2024 | Nature | Peer-reviewed | C3 | G5 | Does not help when outputs are systematically (consistently) wrong |
| 29 | FG-PRM: Fine-grained Hallucination Detection and Mitigation in LM Mathematical Reasoning | Li et al. | 2024 | arXiv | Preprint | C1 | G1 | Data synthesis depends on ground-truth reasoning-step datasets |
| 30 | LLM-Check: Investigating Detection of Hallucinations in LLMs | Sriramanan et al. | 2024 | NeurIPS 2024 | Peer-reviewed | – | – | Detection rates leave room at low false-positive rates |

| # | Title | First author | Year | Venue | Type | Clusters | Gaps | Representative stated limitation |
|---|---|---|---|---|---|---|---|---|
| 31 | Survey on Hallucination in Reasoning LLMs: Evaluation, Taxonomy, Intervention, Open Issues | Liu et al. | 2026 | Data Intelligence | Peer-reviewed | – | – | Closed-form questions only; 50 questions; subjective coding |
| 32 | Mathematical Computation and Reasoning Errors by Large Language Models | Zhang & Graf | 2025 | AIME-Con 2025 | Peer-reviewed | – | – | LLM and human labelling not independent; only three item models |
| 33 | Large Language Models and Mathematical Reasoning Failures | Boye & Moell | 2025 | arXiv | Preprint | C11 | G2 | 50 problems, single query per model; calculus and trigonometry missing |
| 34 | Self-Consistency-Based Hallucination Detection for Mathematical Reasoning* | Liu et al. | 2025 | arXiv | Preprint | C3 | G5 | Misses consistent but incorrect reasoning (”collective hallucination”) |
| 35 | ErrorRadar: Benchmarking Complex Mathematical Reasoning of MLLMs via Error Detection* | Yan et al. | 2026 | Findings of ACL 2026 | Peer-reviewed | C2, C6, C9 | G3, G4, G9 | Static evaluation only; K-12 coverage limited |
| 36 | ProcessBench: Identifying Process Errors in Mathematical Reasoning* | Zheng et al. | 2025 | ACL 2025 | Peer-reviewed | C11 | G2 | Possible inaccurate error-location labels; first error only |
| 37 | Let’s Verify Step by Step (PRM800K)* | Lightman et al. | 2024 | ICLR 2024 | Peer-reviewed | C1, C4, C7, C11 | G1, G2, G8 | Correct-answer, wrong-reasoning solutions misgraded; contamination not ruled out |
| 38 | Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations* | Wang et al. | 2024 | ACL 2024 | Peer-reviewed | C1, C7 | G1, G8 | Impact of automatic-label noise ”still undetermined” |
| 39 | The Lessons of Developing Process Reward Models in Mathematical Reasoning* | Zhang et al. | 2025 | Findings of ACL 2025 | Peer-reviewed | C1, C4, C7 | G1, G2, G8 | MC estimation introduces high label noise; BoN evaluation biased |
| 40 | Step-DPO: Step-wise Preference Optimization for Long-Chain Reasoning | Lai et al. | 2024 | arXiv | Preprint | – | – | First-error localisation needs manual or GPT-4 verification |
| 41 | Multiple Choice Questions: Reasoning Makes LLMs More Self-Confident, Especially When Wrong* | Fu et al. | 2026 | IEEE Intelligent Systems | Peer-reviewed | C3 | G5 | CoT inflates confidence when wrong; causal mechanism open |
| 42 | SMRC: Aligning LLMs with Student Reasoning for Mathematical Error Correction* | Zeng et al. | 2025 | arXiv | Preprint | C2, C6, C9 | G3, G4, G9 | Benchmarks use synthetic student responses; models struggle on authentic reasoning |
| 43 | Right Is Not Enough: Pitfalls of Outcome Supervision in Training LLMs for Math Reasoning | Guo et al. | 2025 | arXiv | Preprint | C1, C4 | G1, G2 | Verifier quality bound to base LLM; costly for largest models |
| 44 | Error Classification of LLMs on Math Word Problems: A Dynamically Adaptive Framework | Sun et al. | 2025 | arXiv | Preprint | C5, C10 | G6, G7 | Excludes correct-answer, flawed-process cases |
| 45 | Error Detection and Correction for Interpretable Mathematics in LLMs (EDCIM) | Yang et al. | 2025 | AAAI Fall Symposium 2025 | Peer-reviewed | C10 | G6 | Manually engineered error rules; symbolic solver scalability |
| 46 | Assessing Logical and Arithmetic Errors across Wide Numerical Ranges | Shrestha et al. | 2025 | arXiv | Preprint | – | – | Small models, GSM8K only; coarse grading |

**Cluster sizes.** C1: n=11 (#6, #7, #8, #9, #15, #19, #29, #37, #38, #39, #43); C2: n=5 (#6, #9, #19, #35, #42); C3: n=5 (#15, #16, #28, #34, #41); C4: n=9 (#1, #2, #3, #4, #5, #25, #37, #39, #43); C5: n=6 (#1, #2, #3, #6, #16, #44); C6: n=5 (#6, #20, #21, #35, #42); C7: n=6 (#3, #4, #15, #37, #38, #39); C8: n=5 (#1, #2, #3, #13, #24); C9: n=5 (#6, #9, #21, #35, #42); C10: n=5 (#5, #11, #17, #44, #45); C11: n=8 (#1, #2, #3, #12, #25, #33, #36, #37).

# S2. External validation log (G1–G10)

Each candidate gap was checked against literature *outside* the corpus in two rounds. **Original check**: June 2026 (G1, G3, G9), 28 September 2026 (G4, G6, G7) and 1 October 2026 (G2, G5, G8, G10); sources and verdicts were recorded, but exact query strings were not. **Logged re-check**: 13 October 2026, with the exact queries listed below, run through a general web search engine; the top results of each query were screened by title and abstract, and decisive papers were opened in full text. Read level: **F** = full text checked on 13 Oct; **A** = abstract or landing page; **C** = corpus paper (five-field extraction); **S** = surfaced in results, not examined further. Statuses: *addressed* (existing work solves the problem as formulated); *partially addressed* (substantial progress in specific settings, broader problem still active); *crowded but unresolved* (many competing approaches, no established solution); *narrowed* (broad formulation no longer defensible, a specific residual question remains); *open*. The re-check did not change any status.

## G1 Oracle-free step verification (C1, n=11) — status: Narrowed

| | |
|---|---|
| Original formulation | No reliable step-level verification without an oracle (reference solution, ground-truth answer or frontier judge). |
| Original check | Novelty Verification Report (June 2026): Google Scholar, Semantic Scholar-indexed sources, arXiv, ACL Anthology; abstract level. Evidence: FreePRM, arXiv:2506.03570 [A]; Hard2Verify, ACL 2026 [A]; Consequence-based utility, arXiv:2602.06291 [A]; Step-wise formal verification, arXiv:2505.20869 [A]; AlphaMath Almost Zero (2024) [A]; MathMistake Checker, arXiv:2503.04291 [A] |
| Re-check queries (13 Oct 2026) | `step-level verification math reasoning without reference answer or ground truth labels 2026` |
| Re-check evidence | FreePRM, arXiv:2506.03570 [F]: step labels derived from final-answer correctness during training; inference input is question + solution only<br>ReProbe, arXiv:2511.06209 [A]: internal-state probes for step verification; labels from a larger LLM or self-supervised<br>Hard2Verify, ACL 2026 [S]: venue confirmed (ACL Anthology 2026.acl-long.1031); content as in June check |
| Status and rationale | **Narrowed**. Oracle-light step verification now exists (outcome-supervised PRMs, probes), all trained and evaluated on model-generated solutions. Verifying a single authentic student solution when the answer is genuinely unknown remains insufficiently addressed. |

## G2 Reasoning-quality evaluation (C4, C11, n=9) — status: Addressed

| | |
|---|---|
| Original formulation | No framework scores intermediate-step validity automatically, without human annotation or oracle step labels. |
| Original check | External check, 1 Oct 2026: arXiv, ACL Anthology; abstract level. Evidence: ReasonEval (corpus #4) [C]; EvalQReason, arXiv:2602.02295 [A]; Survey of step-by-step reasoning-trace evaluation, Findings of EMNLP 2025 [A]; R-PRM, EMNLP 2025 [A] |
| Re-check queries (13 Oct 2026) | `automatic evaluation of reasoning step quality without human annotation mathematics LLM` |
| Re-check evidence | EvalQReason, arXiv:2602.02295 [A]: annotation-free step-level scoring; strong results on math<br>ReasonEval, AAAI 2025 [C]: step validity and redundancy scoring |
| Status and rationale | **Addressed**. Automatic, annotation-free step-quality evaluation exists for model-generated mathematical reasoning; quality varies across domains, but the gap as formulated no longer holds. |

## G3 Personalised feedback (C6, n=5) — status: Narrowed

| | |
|---|---|
| Original formulation | No system adapts correction style or depth to an individual student's error profile. |
| Original check | Novelty Verification Report (June 2026): Google Scholar, Semantic Scholar-indexed sources, arXiv, ACL Anthology; abstract level. Evidence: Teaching according to students' aptitude, arXiv:2511.15163 [A]; LearnLens, arXiv:2507.04295 [A]; MathEDU, arXiv:2505.18056 [A]; Confirming correct, missing the rest, arXiv:2605.16207 [A]; Pedagogical knowledge-augmented prompting (2025) [A] |
| Re-check queries (13 Oct 2026) | `personalized feedback student math errors LLM tutor error profile across sessions` |
| Re-check evidence | MathEDU, arXiv:2505.18056 [A]: feedback on real student problem-solving processes<br>Personalised error-diagnosis agents, arXiv:2502.13789 [S]: surfaced in results; not examined further |
| Status and rationale | **Narrowed**. Persona-, memory- and forgetting-aware mathematics tutoring exists; tracking a student's recurring error types across sessions and adapting correction to them remains insufficiently addressed. |

## G4 Authentic student data (C2, n=5) — status: Narrowed

| | |
|---|---|
| Original formulation | No paper collects or releases authentic step-by-step student solution data. |
| Original check | External check, 28 Sept 2026: arXiv, ACL Anthology, Semantic Scholar-indexed sources; mostly abstract level. Evidence: ScratchMath, arXiv:2603.24961 [A]; EDU-CIRCUIT-HW, arXiv:2602.00095 [A]; Yu et al., arXiv:2603.00895 [A]; DrawEduMath, arXiv:2501.14877 [A]; MathDial (simulated students) [A] |
| Re-check queries (13 Oct 2026) | `dataset real student handwritten math solutions step-level error annotation LLM 2026` |
| Re-check evidence | ScratchMath, arXiv:2603.24961 [F]: 1,720 primary/middle-school solutions; one error cause and one class per solution; no step-level labels<br>Yu et al., arXiv:2603.00895 [F]: ~800 university calculus students; rubric-guided scores and feedback; no step-level error labels; no carry-forward policy; data not yet released<br>Levine et al., arXiv:2605.19043 [A]: handwritten university STEM work graded at rubric-item level |
| Status and rationale | **Narrowed**. Real handwritten student datasets now exist (school level, or university grading). None found labels every error at step level in university calculus solutions. |

## G5 Calibrated confidence (C3, n=5) — status: Narrowed

| | |
|---|---|
| Original formulation | No method yields calibrated confidence under chain-of-thought without oracle access. |
| Original check | External check, 1 Oct 2026: arXiv, ACL Anthology; abstract level. Evidence: Calibration drift under reasoning, arXiv:2606.11211 [A]; Don't think twice, arXiv:2508.15050 [A]; CoT-UQ, Findings of ACL 2025 [A]; Reasoning helps surface self-confidence signals, UncertaiNLP 2025 [A]; When can we trust LLM graders?, arXiv:2603.29559 [A]; CHiL(L)Grader, arXiv:2603.11957 [A] |
| Re-check queries (13 Oct 2026) | `LLM confidence calibration step-level error detection student solutions tutoring` |
| Re-check evidence | Srivatsa et al., EMNLP 2025 [A]: first-error localisation in student-like solutions; no calibration analysis<br>LLMs as students who think aloud, arXiv:2602.01015 [S]: surfaced; not examined further |
| Status and rationale | **Narrowed**. CoT overconfidence is studied and partly mitigated; calibrated graders route uncertain cases to humans, but for short answers. Calibrated step-level error judgements on student mathematics remain open. |

## G6 Multi-error detection (C10, n=5) — status: Narrowed

| | |
|---|---|
| Original formulation | All systems evaluate single-type, single-step errors; entangled multiple errors are excluded. |
| Original check | External check, 28 Sept 2026: arXiv, ACL Anthology, Semantic Scholar-indexed sources; mostly abstract level. Evidence: MWP-MISTAKE, ACL 2025 [A]; ErrorRadar (corpus #35) [C]; ProcessBench (corpus #36) [C]; ScratchMath, arXiv:2603.24961 [A] |
| Re-check queries (13 Oct 2026) | `detecting multiple errors in one math solution LLM benchmark all errors not only first error` |
| Re-check evidence | MWP-MISTAKE, ACL 2025 [F]: rule-based errors (one per variant) and small-LLM errors that propagate across steps; no real student solutions<br>Srivatsa et al., EMNLP 2025 [A]: first error only |
| Status and rationale | **Narrowed**. Multi-step propagating errors are studied only with synthetic or model-generated errors; co-occurring errors in real student solutions remain open. |

## G7 Error propagation (C5, n=6) — status: Narrowed

| | |
|---|---|
| Original formulation | No propagation model exists; systems model only the first error. |
| Original check | External check, 28 Sept 2026: arXiv, ACL Anthology, Semantic Scholar-indexed sources; mostly abstract level. Evidence: ARES (corpus #16), EMNLP 2025 [C]; Beyond the First Error, Findings of EMNLP 2025 [A]; MWP-MISTAKE, ACL 2025 [A]; Srivatsa et al., EMNLP 2025 [A] |
| Re-check queries (13 Oct 2026) | `error propagation reasoning chain process reward model propagated errors after first error`<br>`”error carried forward” LLM grading student mathematics`<br>`LLM automated marking ”follow-through” OR ”consequential error” OR ”carry-forward” partial credit mathematics student answers`<br>`LLM distinguish independent errors from propagated errors student math solution annotation` |
| Re-check evidence | ARES, arXiv:2507.12948 [F]: a step logically correct w.r.t. a wrong premise is scored as a (propagated) error; no separate native/propagated labels; synthetic and LLM chains only<br>Beyond the First Error, arXiv:2505.14391 [F]: ’Error Propagation’: a step building on an earlier error is labelled incorrect; LLM-generated long CoT; o1 judge labels<br>Forest of Errors, arXiv:2604.02967 [A]: error accumulation in reasoning models; no student data<br>arXiv:2603.00451; arXiv:2603.14732 [A]: returned by carry-forward queries; grading studies that do not address carried-forward errors |
| Status and rationale | **Narrowed**. Propagation-aware checking exists, but it counts steps that build on an earlier error as errors, on model-generated chains. No work found tests whether LLMs separate a new error from a step that correctly carries forward an earlier wrong value (teachers’ ’error carried forward’) in student work. |

## G8 Noise-robust process supervision (C7, n=6) — status: Partially addressed

| | |
|---|---|
| Original formulation | Automatic process labels are noisy and no robust remedy exists. |
| Original check | External check, 1 Oct 2026: arXiv, ACL Anthology; abstract level. Evidence: SCAN, NeurIPS 2025 [A]; Noise-aware PRM learning, arXiv:2601.12748 [A] |
| Re-check queries (13 Oct 2026) | `noise-robust process reward model Monte Carlo label noise denoising 2026` |
| Re-check evidence | SCAN, NeurIPS 2025 [A]: self-denoising MC annotation; large ProcessBench gains<br>Noise-aware PRM learning, arXiv:2601.12748 [A]: label correction and iterative training |
| Status and rationale | **Partially addressed**. Label noise is substantially reduced in specific settings; robustness across domains and verification of student work are not shown. |

## G9 Interactive correction (C9, n=5) — status: Crowded, unresolved

| | |
|---|---|
| Original formulation | Benchmarks judge static solutions; no interactive pedagogical correction. |
| Original check | Novelty Verification Report (June 2026): Google Scholar, Semantic Scholar-indexed sources, arXiv, ACL Anthology; abstract level. Evidence: SocraticLM, NeurIPS 2024 [A]; MathDial, Findings of EMNLP 2023 [A]; StratL, ACL 2025 [A]; UCO, arXiv:2511.08873 [A]; BEA 2025 shared task [A]; Scarlatos et al. (2025) [A]; Discerning minds, arXiv:2508.06583 [A]; TreeInstruct, EMNLP 2024 [A] |
| Re-check queries (13 Oct 2026) | `LLM tutor multi-turn math error remediation real students learning gain 2026` |
| Re-check evidence | Petukhova et al., arXiv:2606.21502 [A]: SFT+DPO tutors for math mistake remediation; no learning-gain measurement<br>Bridging the novice-expert gap, NAACL 2024 [S]: surfaced; not examined further |
| Status and rationale | **Crowded, unresolved**. Many competing interactive tutoring approaches and shared tasks; key datasets use simulated students and evaluation rarely measures learning, so effective correction of authentic student work is unresolved. |

## G10 Intrinsic self-verification (C8, n=5) — status: Partially addressed

| | |
|---|---|
| Original formulation | Self-reflection is externally constructed; no autonomous self-verification. |
| Original check | External check, 1 Oct 2026: arXiv, ACL Anthology; abstract level. Evidence: Self-rewarding correction, arXiv:2502.19613 [A]; S3c-Math (corpus #13) [C]; Spontaneous self-correction, arXiv:2506.06923 [A] |
| Re-check queries (13 Oct 2026) | `intrinsic self-verification LLM mathematical reasoning without external feedback 2026` |
| Re-check evidence | Self-rewarding correction, arXiv:2502.19613 [A]: self-verification and correction at inference without external reward<br>ReVISE, arXiv:2502.14565 [S]: surfaced; not examined further |
| Status and rationale | **Partially addressed**. Self-correction without external feedback is demonstrated for the model's own reasoning; reliable verification of another person's (a student's) solution is not shown. |

## Limitations of this log

One logged query was run per gap (four for G7, the gap most central to the conclusions), so the re-check confirms rather than exhausts the literature. Most evidence was read at abstract level; full texts were checked for the papers on which the G4, G6 and G7 statuses depend, and for FreePRM (G1). The statuses were assigned by the first author; an independent second classification of the ten gaps from this evidence is reported in S3.

# S3. Independent second classification of the gap statuses

The second author classified the ten gaps from the documented evidence (the S2 evidence with the first author's statuses removed), without AI tools, using Google Scholar where needed (about one working day). Gaps were shuffled and relabelled A–J. The first author's statuses were assigned before the second classification and were not changed afterwards. **Exact agreement**: 3 of 10 (G2, G3, G10). In two further gaps (G5, G8) the second coder marked two statuses, one of which matched. **Open vs. not open in broad form**: agreement on 8 of 10; G6 was judged open by the second coder, and G5 was undecided. The disagreements were then resolved by discussion between the two coders, who agreed on the first coder's statuses as final (Table 2 of the paper).

| Gap | Label | Name | First coder | Second coder | Exact match |
|---|---|---|---|---|---|
| G1 | E | Oracle-free step verification | Narrowed | Partially addressed | no |
| G2 | B | Reasoning-quality evaluation | Addressed | Addressed | yes |
| G3 | I | Personalised feedback | Narrowed | Narrowed | yes |
| G4 | G | Authentic student data | Narrowed | Addressed | no |
| G5 | C | Calibrated confidence | Narrowed | Narrowed or Open (two marked) | partly |
| G6 | J | Multi-error detection | Narrowed | Open | no |
| G7 | A | Error propagation | Narrowed | Partially addressed | no |
| G8 | F | Noise-robust process supervision | Partially addressed | Addressed or Partially addressed (inclined to Addressed) | partly |
| G9 | D | Interactive correction | Crowded but unresolved | Partially addressed | no |
| G10 | H | Intrinsic self-verification | Partially addressed | Partially addressed | yes |